

An analytical solution to remember

If there is only one analytical solution to an ODE that you remember, it should be this one. You *must* memorize it!

Many different systems that you might try to model end up with an ODE that has this form:

$$\frac{dx}{dt} = k \cdot x$$

where x is the state variable, t is time, and k is a rate constant.

It is a linear, first-order ODE.

Where does it show up? All over! For example:

1. In first-order chemical reactions, where x might be a the concentration or mass of a reactant and k is entered as the negative of the first-order rate constant. Note, confusingly, that "first-order" here is different from a first-order ODE.
2. For unlimited growth of bacteria or any other organism, where x is microbial biomass and k is the intrinsic rate of growth.
3. In heat transfer by a lumped capacitance method, where x is temperature, k has a negative sign and includes a convection heat transfer coefficient, specific heat capacity, probably exposed surface area and object mass as well, and possibly other sources of surface or near-surface resistance.

So you really do need to be able to solve this type of ODE. Below I'll go through the steps in solving it, but ultimately you will probably find it easiest to just memorize the solution. Remember you can always check a solution as shown below.

$$\frac{dx}{dt} = kx$$

Separation of variables

$$\frac{1}{x} dx = k dt$$

Integrate both sides

$$\int \frac{1}{x} dx = \int k dt$$

You could look up these integrals in a table if needed (at least the RHS should be very easy!)

$$\ln|x| + c_1 = kt + c_2$$

We can combine the constants of integration

$$\ln|x| = kt + C$$

Then exponentiate (take the antilog, or $\exp()$, of each side).

$$|x| = e^{kt + C}$$

Remember the exponent rules?

$$|x| = e^C \cdot e^{kt}$$

A

We can call the new constant "A", which can also get the +/- that comes from the $|x|$.

$$x = A \cdot e^{kt}$$

And that is the general solution. The value of A is the value of x at time 0 (because $\exp(0) = 1$).
So we could write

$$\boxed{x = x_0 \cdot e^{kt}}$$

(1)

We can check this solution and the solution for any ODE by differentiating and seeing if it indeeds matches the original ODE. You can check a table of derivatives to see that the derivative of any $a \cdot \exp(b \cdot t)$ with respect to t is just $b \cdot a \cdot \exp(b \cdot t)$. So

$$\frac{d}{dt} (x_0 \cdot e^{kt}) = k \cdot x_0 \cdot e^{kt}$$

$\parallel$
 x

And recognizing the part of the solution that is x (from equation 1 above), we see that in fact

$$\frac{dx}{dt} = kx$$

So that proves that equation 1 is a solution to the original ODE.